

COSMO Cube

COSMO® Cube

Manufacturing & QA Manual

LOT Systems, Inc. — COSMO® CIA Hardware Division

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Classification: Production · Target: 100-unit run · Inventor: Vadim Marmeladov

COSMO® Cube — Manufacturing Guide v1.0

Document: COSMO-MANUFACTURING-v1.md

Target: 100-unit production run

Manufacturer: PCBWay (primary)

Date: 2026-06-12

1. Manufacturing Overview

The COSMO® Cube is produced across three parallel manufacturing streams:

Stream A: PCBWay Electronics

- └─ PCB fabrication (4-layer, 35×35mm)
- └─ SMT component placement (turnkey)
- └─ Reflow soldering (lead-free SAC305)
- └─ Post-reflow AOI inspection

Stream B: PCBWay CNC Machining

- └─ 316L SS back plate (mirror polished)
- └─ 316L SS front bezel (satin + cutouts)
- └─ CNC tolerances $\pm 0.05\text{mm}$
- └─ Laser engraving

Stream C: Custom Components

- └─ LiPo battery (Grepow, 35×35×2.5mm)
- └─ OLED display modules
- └─ Camera modules (Himax HM01B0)
- └─ Wireless charging coils

Final Assembly (USA or designated facility):

- └─ PCB → coil → battery → enclosure stacking
- └─ Firmware flashing + provisioning
- └─ QA testing (every unit)
- └─ Packaging

2. PCBWay Order — Step by Step

2.1 PCB Fabrication Order

URL: <https://www.pcbway.com/orderonline.aspx>

Files to upload: - Gerber files (ZIP): COSMO-PCB-v1-gerbers.zip - Copper layers: F.Cu, B.Cu, In1.Cu, In2.Cu - Drill files: drill.drl, drill-NPTH.drl - Paste layers: F.Paste, B.Paste - Silk: F.Silkscreen - Mask: F.Mask, B.Mask - Outline: Edge.Cuts

PCBWay Order Parameters:

Layers:	4
Board Dimensions:	35 × 35 mm
Board Thickness:	0.8 mm
Quantity:	110
Surface Finish:	ENIG
Copper Weight:	1 oz (outer), 0.5 oz (inner)
Min Hole Size:	0.2 mm (laser via)
Min Track/Space:	4/4 mil
Solder Mask:	Black
Silkscreen:	White (top only)
Via Filling:	Filled + capped (IPC-4761 Type VII)
IPC Class:	Class 2
RoHS:	Yes

Estimated price (110 boards, 4-layer): \$280–\$380

2.2 SMT Assembly Order (Turnkey)

URL: <https://www.pcbway.com/assembly/>

Files to upload: - BOM file: COSM0-BOM-Assembly.csv (see format below) -
Pick-and-Place file: COSM0-CPL-v1.csv - Schematic PDF: COSM0-Schematic-v1.pdf

BOM CSV Format for PCBWay:

```
Comment,Designator,Footprint,Quantity,LCSC/Mouser  
MPN,Supplier,Notes  
ESP32-S3-MINI-1U-N8,U1,ESP32S3-MINI-1U,1,713-ESP32-S3-MINI-  
1U,Mouser,Provide socket  
SSD1327 OLED,DISP1,FPC-24P-0.5mm,1,SEE BOM,BuyDisplay,DNP - hand  
install  
BME280,U2,BME280-LGA,1,828-BME280,Mouser,Near mesh cutout  
ICM-42688-P,U3,QFN-14-2.5x2.5,1,ICM-42688-P,Mouser,SPI interface  
APDS-9960,U4,LCC-6,1,630-APDS-9960,Mouser,Behind display  
BQ51013B,U5,VQFN-20,1,595-BQ51013BRHLR,Mouser,  
BQ25892,U6,WQFN-24,1,BQ25892RTWR,Mouser,  
AP2112K-1.8V,U7,SOT-23-5,1,AP2112K-1.8TRG1,DigiKey,  
W25Q64JV,U8,SOP-8,1,W25Q64JVSSIQ,DigiKey,Optional  
RGB LED,LED1,1206,1,APTR3216ZGCK,Mouser,  
Tactile Button,SW1,SMD-6x6mm,1,EVQ-Q2C03W,Mouser,
```

100nF cap, C1-C20, 0402, 20, GCM155R71C104KA55D, Mouser, Decoupling
10µF cap, C21-C25, 0603, 5, GRM188R61A106KE69D, Mouser, Bulk
10kΩ res, R1-R10, 0402, 10, RC0402FR-0710KL, Mouser, Pull-up/down

Assembly Parameters:

Solder paste: SAC305 (lead-free)
Reflow: Standard lead-free profile (peak 260°C)
Testing: 100% AOI
X-ray: On U5 (BQ51013B VQFN), U6 (BQ25892 WQFN)
DNP list: DISP1, J_CAM (camera connector), J_BAT (battery connector)

Estimated assembly cost (110 units): \$900–\$1,500

2.3 CNC Machining Order

URL: <https://www.pcbway.com/rapid-prototyping/manufacture/?type=cnc>

Files to upload: - Part A (Back plate): COSMO-Back-Plate.step + COSMO-Back-Plate.dxf - Part B (Front bezel): COSMO-Front-Bezel.step + COSMO-Front-Bezel.dxf - Technical drawing: COSMO-Enclosure-Drawing-v1.pdf

Part A — Back Plate Specifications:

Material: 316L Stainless Steel
Dimensions: 40.0 × 40.0 × 0.8 mm (overall)
Features:

- Flat plate, no through-holes (wireless charging only)
- 4× M1.0 threaded blind holes, 1.5mm deep, at corners (3mm from each edge)
- Stepped rim: 0.3mm × 0.3mm rebate around perimeter for gasket

Surface:

- Face finish: Mirror polished #8 (Ra <0.025µm)
- Rim finish: Bead blasted satin

Engraving:

- "LOT@" center, 8mm wide, 0.2mm deep laser etch
- "COSMO@ CIA" lower-right, 4mm wide, 0.15mm deep
- Serial number lower-left, 3mm wide (variable per unit)

- Fill all engravings with black epoxy
Qty: 110

Part B — Front Bezel Specifications:

Material: 316L Stainless Steel
Overall: 40.0 × 40.0 × 4.2 mm (assembled height)
Wall: 0.5mm minimum thickness
Features:
- Rectangular frame with 5mm rim
- Central display aperture: 29 × 29mm (glass/OLED seats inside)
- 4× M1.0 countersunk through-holes at corners
- Camera aperture: 5mm diameter circle, bottom-left (8mm from left, 5mm from bottom)
- Button aperture: 8mm diameter circle, bottom-center, 4mm depth recess
- LED aperture: 2.5mm diameter circle, bottom-right
- Weather mesh: 5×5mm area, 0.5mm hole grid, right-center
- EPDM gasket channel: 0.5 × 0.5mm square groove, inner perimeter
- PCB standoffs: 4× M1.0 × 1.5mm blind post, interior, 5mm from corners
Surface:
- Exterior: Bead blasted satin (Ra 0.4–1.6µm)
- Interior: As-machined (no finish)
Qty: 110

CNC Order Notes for PCBWay: > “We require 316L (not 304) stainless steel for corrosion resistance. > Back plates require mirror polishing to #8 finish — Ra <0.025µm. > Front bezels require bead blast satin finish. > All laser engraving to be filled with black epoxy (standard jewelry technique). > Serial numbers are sequential: CQ-001-26 through CQ-110-26. > Please provide per-unit serial engraving as a variable field. > Requested dimensional tolerance: ±0.05mm. > Request DFM review before production start.”

Estimated CNC cost (110 sets): \$4,000–\$6,000

3. Quality Control

3.1 PCB QA Checkpoints

Test	Method	Pass Criteria
Visual inspection	AOI (automated)	No bridges, missing parts, tombstones
Solder joint quality	X-ray (BQ chips)	Full pad coverage, no voids >25%
Power rail voltage	Bench measurement	3.3V $\pm$ 2%, 1.8V $\pm$ 2%
WiFi connectivity	ESP-IDF test firmware	-70dBm or better at 1m
Display function	Firmware test screen	All 16 gray levels displayed
Button function	Bench press + scope	Clean falling edge, < 5ms bounce
Sensor readings	Firmware sensor test	Temp $\pm$ 2°C of reference, humidity $\pm$ 5%
Wireless charging	Qi pad @ 5W	Full charge current within 30s of placement

3.2 Enclosure QA

Test	Method	Pass Criteria
Dimension check	Caliper measurement	$\pm$ 0.1mm on all critical dims
Mirror finish	Visual + Ra measurement	No scratches, Ra <0.05 μ m
Engraving depth	Optical profilometer	0.15–0.25mm depth
Assembly fit	Hand assembly test	No gap > 0.1mm when assembled
IP54 splash	5-min water spray test	No water ingress

3.3 System QA (Every Unit)

COSMO® Cube QA Checklist – Unit: CQ-____-26

- ☐ 1. Flash firmware v1.0.0-001
- ☐ 2. Run provisioning tool – assign serial + API key
- ☐ 3. Boot – verify boot screen displays
- ☐ 4. WiFi connection – verify connects within 30s
- ☐ 5. API poll – verify notification received from LOT test server

- ☐ 6. Notification display – verify text renders correctly on OLED
- ☐ 7. Copy button – verify LED pulse + Log entry on lot-systems.com
- ☐ 8. BME280 – verify temp/humidity/pressure within calibrated range
- ☐ 9. APDS-9960 – verify ambient light value, proximity response
- ☐ 10. ICM-42688 – verify accelerometer reads ≈ 9.81 Z-axis
- ☐ 11. Wireless charging – verify charge current, LED indicator
- ☐ 12. Deep sleep – verify wakeup from button press
- ☐ 13. Enclosure assembly – secure M1.0 screws (torque: 0.3 N·m)
- ☐ 14. Final visual – no scratches on mirror face, clean front

QA Technician: _____ Date: _____ Pass / Fail

4. Production Timeline

4.1 Gantt Chart (Target: 10-Week Build)

Week	Activity
1	PCB design finalized → Gerbers generated Order: Battery (Grepow) – 8-week lead Order: SS enclosure (PCBWay CNC) – 3-week lead
2	Order: All SMD components (Mouser/DigiKey) Order: OLED displays (BuyDisplay) – 2-week lead Order: Camera modules (ArduCam) Order: Qi coils + ferrite (Alibaba)
3	PCB + SMT order placed (PCBWay) – 2-week lead Firmware development begins (ESP-IDF)
4	Components arrive (SMD, sensors) Firmware: WiFi + API integration
5	PCB boards arrive (PCBWay) Prototype assembly (10 units) Firmware: Display driver, button handler
6	Prototype QA – identify issues CNC enclosures arrive (PCBWay CNC) Firmware: Sensor drivers, power management

```
7 | Firmware: OTA, security, factory provisioning
  | LOT backend: hardware API endpoints
  |
8 | Battery arrives (Grepow)
  | Full production assembly begins (100 units)
  | LOT Log tab: hardware entry display
  |
9 | QA testing – all 100 units
  | Failed units: rework or replace
  | Packaging
  |
10 | 100-unit build complete
   | Documentation finalized (PDF export)
   | Ship to Vadim / warehouse
```

5. Engineering Files Required

5.1 For PCBWay PCB Order

```
cosmo-pcb-v1/
├── gerbers/
│   ├── COSM0-F_Cu.gbr          # Front copper
│   ├── COSM0-B_Cu.gbr          # Back copper
│   ├── COSM0-In1_Cu.gbr        # Inner layer 1 (GND plane)
│   ├── COSM0-In2_Cu.gbr        # Inner layer 2 (Power plane)
│   ├── COSM0-F_Mask.gbr
│   ├── COSM0-B_Mask.gbr
│   ├── COSM0-F_Paste.gbr
│   ├── COSM0-F_Silkscreen.gbr
│   ├── COSM0-Edge_Cuts.gbr
│   ├── COSM0-drill.drl
│   └── COSM0-drill-NPTH.drl
├── COSM0-BOM-Assembly.csv      # BOM for turnkey SMT
├── COSM0-CPL-v1.csv           # Pick and place coordinates
└── COSM0-Schematic-v1.pdf
```

5.2 For PCBWay CNC Order

```
cosmo-enclosure-v1/
├── 3D-Models/
```

```

|   └─ COSM0-Back-Plate.step
|   └─ COSM0-Front-Bezel.step
└─ 2D-Drawings/
    └─ COSM0-Back-Plate.dxf
    └─ COSM0-Front-Bezel.dxf
    └─ COSM0-Enclosure-Drawing-v1.pdf
└─ Engraving/
    └─ COSM0-Engraving-Layout.ai # Adobe Illustrator for laser
paths

```

Design tool recommendation: KiCad 8.0 (PCB, free, open-source), FreeCAD or Fusion 360 (enclosure CAD).

6. Cost Summary

Item	Unit Cost	100 Units	110 Ordered
PCB fabrication	\$3.50	\$350	\$385
SMT assembly	\$12.00	\$1,200	\$1,320
ESP32-S3-MINI-1U	\$3.80	\$380	\$418
OLED display	\$5.50	\$550	\$605
Camera (HM01B0)	\$4.20	\$420	\$462
BME280	\$2.80	\$280	\$308
ICM-42688-P	\$3.50	\$350	\$385
APDS-9960	\$2.10	\$210	\$231
Qi Rx IC (BQ51013B)	\$2.80	\$280	\$308
Qi Rx coil + ferrite	\$1.50	\$150	\$165
Power IC (BQ25892)	\$2.60	\$260	\$286
Battery (custom LiPo)	\$6.50	\$650	\$715
Passives + misc SMD	\$3.00	\$300	\$330
SS Enclosure (2 parts)	\$40.00	\$4,000	\$4,400
Qi Tx pad (charger)	\$9.00	\$900	—
Packaging	\$4.00	\$400	—

Item	Unit Cost	100 Units	110 Ordered
Subtotal		\$10,730	
Contingency 15%		\$1,610	
Total		\$12,340	
Per-unit cost	\$110		

Target retail price: \$349 (2.9× markup, reflects premium SS finish + LOT platform value)

Bill of Materials

COSMO® Cube — Bill of Materials v1.0

Document: COSMO-BOM-v1.md
Revision: 1.0
Date: 2026-06-12
Quantity: 100 units (+ 10% overage = 110 sets ordered)
Prepared by: Vadim Marmeladov, Inventor

Purchasing Summary

Category	Est. Unit Cost	Est. 100-Unit Total
Main MCU Module	\$4.20	\$420
Display	\$5.50	\$550
Camera	\$4.80	\$480
Weather Sensor	\$3.10	\$310
IMU Sensor	\$3.80	\$380
Ambient Light Sensor	\$2.40	\$240
Wireless Charging (Rx)	\$5.90	\$590
Battery	\$6.50	\$650
Power Management IC	\$2.80	\$280
Passives + Misc SMD	\$3.00	\$300
PCB (4-layer, PCBWay)	\$4.50	\$450
SMT Assembly (PCBWay)	\$12.00	\$1,200
SS Enclosure (2 parts, CNC)	\$38.00	\$3,800
Wireless Charger (Tx pad)	\$9.00	\$900
Packaging	\$4.00	\$400
Total per unit	~\$110	—
100-unit run total	—	~\$10,750
Contingency (15%)	—	\$1,613

Category	Est. Unit Cost	Est. 100-Unit Total
Grand Total	—	~\$12,363

Section 1 — Main Processing

1.1 MCU Module: ESP32-S3-MINI-1U

Field	Detail
Manufacturer	Espressif Systems
MPN	ESP32-S3-MINI-1U-N8
Description	WiFi + BLE5 SoM, 8MB Flash, U.FL antenna, 15.4×20.5mm
Quantity per unit	1
Total ordered	110
Unit price (100+)	~\$3.50–\$4.20
Primary supplier	Mouser Electronics
Mouser MPN	713-ESP32-S3-MINI-1U
Backup supplier	DigiKey — search MPN ESP32-S3-MINI-1U-N8
Datasheet	https://www.espressif.com/sites/default/files/documentation/esp32-s3-mini-1_mini-1u_datasheet_en.pdf
PCBWay note	Specify U.FL variant for external antenna trace on PCB edge

1.2 External SPI Flash (Optional Expansion)

Field	Detail
Manufacturer	Winbond
MPN	W25Q64JVSSIQ
Description	64Mb SPI NOR Flash, SOP-8
Quantity per unit	1 (optional, if 8MB insufficient)
Unit price	~\$0.80
Supplier	DigiKey — search w25q64jvssiq

Section 2 — Display

2.1 OLED Display Module: SSD1327 1.0” 128×128

Field	Detail
Manufacturer	Various (Solomon Systech controller, generic module)
MPN	ER-OLED013-1W (or equivalent 1.0” SSD1327 square module)
Description	1.0” 128×128 grayscale OLED, SPI, 3.3V, white
Quantity per unit	1
Total ordered	110
Unit price (100+)	~\$4.50–\$6.00
Primary supplier	Buydisplay.com or BuyDisplay.cn
Backup supplier	Alibaba — search 1 inch SSD1327 OLED 128x128
Alternate MPN	Adafruit 1431 (for prototyping)
Notes	Specify SPI interface (not I2C) for speed. Confirm 3.3V logic. Order with 0.5mm FPC connector.

Section 3 — Camera

3.1 Camera Module: Himax HM01B0

Field	Detail
Manufacturer	Himax Technologies
MPN	HM01B0
Description	Ultra-low power 320×320 CMOS camera, DVP, CSP package
Quantity per unit	1

Field	Detail
Total ordered	110
Unit price (100+)	~\$3.50–\$5.00
Primary supplier	ArduCam (arducam.com) — sells HM01B0 breakout/module
Alternate supplier	Himax direct via distributor — search HM01B0 - AAA
Datasheet	Available via Himax NDR program
Lens	Requires M7 macro lens, 2.35mm EFL, HFOV 66°
Notes	Request with included M7 lens module for thin mounting. Total camera + lens stack: ~3.5mm

Fallback camera (prototype/simpler):

Field	Detail
MPN	OV2640 (2MP)
Module	ArduCam Mini 2MP OV2640
Supplier	arducam.com — search OV2640
Notes	Thicker (8mm with lens) — for prototype only, not final 5mm form factor

Section 4 — Environmental Sensors

4.1 Weather Sensor: Bosch BME280

Field	Detail
Manufacturer	Bosch Sensortec
MPN	BME280
Description	Temp/Humidity/Pressure, LGA 2.5×2.5×0.93mm

Field	Detail
Quantity per unit	1
Total ordered	110
Unit price (100+)	~\$2.50–\$3.50
Primary supplier	Mouser Electronics — search BME280
Mouser MPN	828-BME280
Backup supplier	DigiKey — search BME280
Interface	I2C (0x76 or 0x77 address)
Datasheet	https://www.bosch-sensortec.com/products/environmental-sensors/humidity-sensors-bme280/
Notes	Place near weather mesh cutout on PCB for accurate ambient readings. Keep away from battery thermal zone.

4.2 AI-Grade IMU: TDK InvenSense ICM-42688-P

Field	Detail
Manufacturer	TDK InvenSense
MPN	ICM-42688-P
Description	6-axis IMU (accel + gyro), QFN 2.5×2.5×0.91mm
Quantity per unit	1
Total ordered	110
Unit price (100+)	~\$3.20–\$4.50
Primary supplier	Mouser — search ICM-42688-P
Backup supplier	DigiKey — search ICM-42688-P
Interface	SPI (4-wire, up to 24MHz)
Key feature	FIFO 4KB, DMP (hardware motion detection), tap detection, step counter
Notes	Use SPI for speed. Enable FIFO tap detection for button-less gesture.

4.3 Ambient Light + Gesture: Broadcom APDS-9960

Field	Detail
Manufacturer	Broadcom / Avago
MPN	APDS-9960
Description	Gesture + Proximity + ALS + RGBC, LCC 3.94×2.36mm
Quantity per unit	1
Total ordered	110
Unit price (100+)	~\$1.80–\$2.80
Primary supplier	Mouser — search APDS-9960
Mouser MPN	630-APDS-9960
Interface	I2C (address 0x39)
Notes	Mount behind 0.5mm aperture in SS face near display. Used for sleep/wake and notification dismiss gesture.

Section 5 — Power System

5.1 Battery: Custom LiPo 280mAh

Field	Detail
Chemistry	Lithium Polymer
Capacity	280mAh (range: 250–300mAh acceptable)
Dimensions	35mm × 35mm × 2.5mm (H × W × T)
Voltage	3.7V nominal, 4.2V max
Connector	JST-PH 1.25mm 2-pin
Protection	Built-in PCM (overcharge, overdischarge, short)
Quantity per unit	1
Total ordered	110

Field	Detail
Unit price (100+)	~\$5.00–\$8.00
Primary supplier	Shenzhen Grepow Battery — custom cell order: https://www.grepow.com
Backup supplier	Alibaba — search 3535250 LiPo battery 280mAh
Certification	UN38.3 transport certification required
Notes	Request custom order for 35×35×2.5mm. Minimum order may be 200 units — order 200 for buffer.

5.2 Wireless Charging Receiver IC: TI BQ51013B

Field	Detail
Manufacturer	Texas Instruments
MPN	BQ51013BRHLR
Description	Qi-certified wireless power receiver, 5W, VQFN-20
Quantity per unit	1
Total ordered	110
Unit price (100+)	~\$2.50–\$3.50
Primary supplier	Mouser — search BQ51013B
Mouser MPN	595-BQ51013BRHLR
Datasheet	https://www.ti.com/product/BQ51013B
Notes	Requires 30mm Qi receiver coil + resonant capacitors. See reference design BQ51013B EVM.

5.3 Wireless Charging Coil (Receiver)

Field	Detail
Type	Qi receiver coil, 30mm diameter
Inductance	15μH ± 20%

Field	Detail
DCR	< 450mΩ
Thickness	0.4mm (coil only, no ferrite)
Ferrite sheet	35mm × 35mm × 0.3mm (between coil and battery)
Quantity per unit	1 coil + 1 ferrite
Total ordered	110 sets
Unit price	~\$1.00–\$1.80
Supplier	Alibaba — search Qi receiver coil 30mm 15uH or Würth Elektronik 760308101
Notes	Ferrite sheet mandatory to prevent battery RF heating. Stack order: Back SS plate → ferrite → coil → battery → PCB → front SS bezel

5.4 Power Management IC: TI BQ25892

Field	Detail
Manufacturer	Texas Instruments
MPN	BQ25892RTWR
Description	I2C single-cell LiPo charger + OTG + PMIC, WQFN-24
Quantity per unit	1
Total ordered	110
Unit price (100+)	~\$2.20–\$3.20
Primary supplier	Mouser — search BQ25892
Datasheet	https://www.ti.com/product/BQ25892
Notes	Handles Qi input, charges LiPo, outputs regulated 3.3V system rail. I2C configuration by ESP32-S3.

5.5 LDO Regulator (1.8V for camera): Diodes Inc AP2112K-1.8TRG1

Field	Detail
MPN	AP2112K-1.8TRG1
Description	600mA LDO, 1.8V output, SOT-23-5
Quantity per unit	1
Unit price	~\$0.30
Supplier	DigiKey — search AP2112K-1.8TRG1

Section 6 — User Interface

6.1 Copy Button: Stainless Steel Tactile Dome

Field	Detail
Type	Stainless steel tactile dome, SMD
Diameter	8mm (external cap)
Actuation Force	180gF (firm, intentional press)
Travel	0.2mm
Life	1,000,000 cycles
MPN	Panasonic EVQ-Q2C03W or equivalent
Quantity per unit	1
Unit price	~\$0.25
Supplier	Mouser — search EVQ-Q2C03W
Notes	SS cap machined into front bezel as integrated dome. Button signal to ESP32-S3 GPIO with hardware debounce (100nF + 10kΩ).

6.2 Status LED: RGB LED, 1206 SMD

Field	Detail
MPN	Kingbright APTR3216ZGCK (or Cree CLVBA-FKA)

Field	Detail
Description	RGB LED, 1206, common cathode
Quantity per unit	1
Unit price	~\$0.20
Supplier	Mouser or DigiKey — search APTR3216ZGCK
Notes	Behind 0.8mm frosted SS diffuser cutout. PWM-controlled by ESP32-S3 (GPIO).

Section 7 — PCB

7.1 Custom 4-Layer PCB

Field	Detail
Manufacturer	PCBWay
Service URL	https://www.pcbway.com/orderonline.aspx
Layer Count	4
Board Size	35mm × 35mm
Board Thickness	0.8mm
Copper Weight	1oz outer, 0.5oz inner
Surface Finish	ENIG (Electroless Nickel Immersion Gold)
Solder Mask	Black
Silkscreen	White (front side only)
Min Trace/Space	4mil / 4mil
Min Drill	0.2mm (laser via)
Via Fill	Filled + capped vias for flat surface
Quantity	110 boards
Est. Price	~\$180–\$350 total (100–110 units, 4-layer)
PCBWay Order	Upload Gerbers at https://www.pcbway.com/orderonline.aspx

Field	Detail
Delivery	5–7 business days (standard), 2–3 days (rush)
Notes	Use IPC Class 2. Request DFM review. Specify RoHS-compliant materials.

7.2 SMT Assembly (Turnkey)

Field	Detail
Service	PCBWay Turnkey SMT Assembly
Service URL	https://www.pcbway.com/assembly/
Type	Full turnkey (PCBWay sources components from BOM)
Solder Paste	SAC305 lead-free (RoHS)
Reflow Profile	Standard SMD lead-free profile
Testing	100% AOI, optional X-ray for BGA
Quantity	110 boards assembled
Est. Assembly Cost	~\$800–\$1,400 total
Files required	BOM (this document), Gerbers, Pick-and-Place (CPL), schematic
Notes	Mark ESP32-S3-MINI-1U, display connector, camera connector as DNP sockets — hand-assemble after reflow. Mark battery connector as DNP — connect during final assembly.

Section 8 — Mechanical Enclosure

8.1 Stainless Steel Enclosure (2-Part)

Field	Detail
Material	316L Stainless Steel

Field	Detail
Parts	2: (A) Back plate + (B) Front bezel
Manufacturer	PCBWay CNC Service
CNC URL	https://www.pcbway.com/rapid-prototyping/manufacture/?type=cnc
External Dimensions	40mm × 40mm × 5mm
Wall Thickness	0.4mm minimum (back plate), 0.5mm bezel walls
Assembly Method	4× M1.0 countersunk screws + continuous gasket seal
Tolerances	±0.05mm (CNC machined)
Back Plate Finish	Mirror polish #8 (Ra <0.025µm) — specify “mirror polishing” in PCBWay order
Front Bezel Finish	Bead-blasted satin (Ra 0.4–1.6µm)
Engraving	Laser-etched, 0.2–0.3mm depth, filled with black epoxy
Gasket	EPDM O-ring, 38mm × 38mm square, 0.5mm cross-section
Quantity	110 sets (220 pieces total)
Unit price (100 sets)	~\$32–\$45 per set (CNC + finishing at 100 qty)
Total cost	~\$3,500–\$4,500
Lead Time	7–14 business days
Notes	Request separate quotes for: (1) raw CNC, (2) mirror polishing (back), (3) bead blast (front), (4) laser engraving. PCBWay CNC team handles all. Attach 2D drawings (DXF) + 3D model (STEP).

8.2 Internal Screws

Field	Detail
Type	M1.0 × 2.0mm flat-head, SS316

Field	Detail
Quantity per unit	4
Supplier	Alibaba — search M1.0 stainless steel screw flat head
Unit price	~\$0.05 each

8.3 Thermal Interface Material

Field	Detail
Type	Thermal pad, 1mm thick, 6 W/m·K
Size	20mm × 20mm (cut to fit over ESP32-S3 module)
MPN	Fujipoly XR-m (or equivalent)
Notes	Between ESP32-S3 and SS enclosure for passive heat transfer

Section 9 — Wireless Charger (Transmitter)

9.1 Desktop Qi Charging Pad (for user kit)

Field	Detail
Standard	Qi 5W
Form Factor	Square pad, 80mm × 80mm × 6mm
Input	USB-C, 5V/2A
Finish	Matte black TPU top, SS bottom weight
LED Indicator	Breathing white LED during charge
Cable	1.5m USB-C to USB-A, braided
Brand/Supplier	OEM Qi pad — Alibaba search 5W Qi charging pad square OEM
Quantity	100 units (1 per device)
Unit price (100+)	~\$6.00–\$10.00
LOT Branding	Request custom pad with LOT® wordmark silk-printed on top

Field	Detail
Notes	Specify Qi-certified transmitter IC (TI BQ500215 or STWBC series). Request CE/FCC pre-certified module.

Section 10 — Packaging

10.1 Retail Box

Field	Detail
Type	Rigid gift box, magnetic closure
Dimensions	100mm × 100mm × 60mm
Material	Matte black rigid cardboard, spot UV on LOT® logo
Interior	Custom foam tray (EVA foam, cut-to-fit)
Contents per box	(1) COSMO® Cube, (1) Qi charging pad + cable, (1) Quick Start card, (1) QR code card → firmware docs
Quantity	100 boxes
Supplier	Alibaba — search custom rigid gift box magnetic 100x100x60
Unit price (100+)	~\$3.00–\$5.00

10.2 Documentation Insert

Field	Detail
Format	4-page folded card, 100mm × 100mm
Content	Device overview, pairing instructions, button function, Qi charging, support QR code
Print	4-color CMYK, 300gsm uncoated
Printer	Local US print shop or Vistaprint for 100 qty

Section 11 — Component Purchasing Order

Priority Order Sequence

Week 1 (Order immediately – longest lead time):

- ☐ Custom LiPo battery (Grepow, 6-8 week lead)
- ☐ SS enclosure quote + order (PCBWay CNC, 2-3 week)
- ☐ Qi receiver coils + ferrite (Alibaba, 2-3 week shipping)

Week 2 (Order in parallel):

- ☐ ESP32-S3-MINI-1U (Mouser, 1-2 week)
- ☐ BME280 (Mouser, in-stock)
- ☐ ICM-42688-P (Mouser, 1-2 week)
- ☐ APDS-9960 (Mouser, in-stock)
- ☐ BQ51013B (Mouser, 1-2 week)
- ☐ BQ25892 (Mouser, 1-2 week)
- ☐ OLED display modules (BuyDisplay, 2-3 week)

Week 3 (After PCB Gerbers ready):

- ☐ Submit PCB + SMT order to PCBWay (2-3 week turnkey)
- ☐ Camera modules (ArduCam, 1-2 week)
- ☐ Packaging (Alibaba, 3-4 week)

Week 4-5:

- ☐ PCB boards arrive
- ☐ Begin enclosure assembly
- ☐ Firmware flash + QA

Week 6:

- ☐ 100-unit build complete

Section 12 — Key Supplier Links

Supplier	Use	URL
PCBWay PCB	PCB fabrication	https://www.pcbway.com/orderonline.aspx
PCBWay Assembly	SMT turnkey	https://www.pcbway.com/assembly/

Supplier	Use	URL
PCBWay CNC	SS enclosure machining	https://www.pcbway.com/rapid-prototyping/manufacture/?type=cnc
Mouser Electronics	Electronic components	https://www.mouser.com
DigiKey	Electronic components	https://www.digikey.com
ArduCam	Camera modules	https://www.arducam.com
Grepow Battery	Custom LiPo	https://www.grepow.com
BuyDisplay	OLED displays	https://www.buydisplay.com
Texas Instruments	BQ51013B, BQ25892 datasheets	https://www.ti.com
Bosch Sensortec	BME280	https://www.bosch-sensortec.com
Espressif	ESP32-S3	https://www.espressif.com

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Inventor: Vadim Marmeladov — 2026-06-12